Department of Mathematics
IIT Hyderabad
MA1130: Vector Calculus
Credits: 1
Fall 2026
Assignment 1

1. Prove or give a counterexample for each of the following statements:
   (a) If $\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{w}$, then $\mathbf{v} = \mathbf{w}$.
   (b) If $\mathbf{v} \cdot \mathbf{w} = 0$ for all $\mathbf{v}$, then $\mathbf{w} = \mathbf{0}$.
   (c) If $\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{w}$ for all $\mathbf{u}$, then $\mathbf{v} = \mathbf{w}$.

2. Prove that
$$\big|\, \|\mathbf{v}\| - \|\mathbf{w}\| \,\big| \leq \|\mathbf{v} - \mathbf{w}\|$$
for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$.

3. Calculate the area of the triangle $\triangle PQR$, where
$$P = (5, 1, -2), \quad Q = (4, -4, 3), \quad R = (2, 4, 0).$$

4. Calculate the area of the parallelogram $PQRS$, where
$$P = (-2, -2), \quad Q = (1, 4), \quad R = (6, 6), \quad S = (3, 0).$$

5. If $\mathbf{v}$ and $\mathbf{w}$ are unit vectors in $\mathbb{R}^3$, under what condition(s) would $\mathbf{v} \times \mathbf{w}$ also be a unit vector in $\mathbb{R}^3$? Justify your answer.

6. Prove the following statements for all vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$:
   (a) $\|\mathbf{v} \times \mathbf{w}\|^2 + |\mathbf{v} \cdot \mathbf{w}|^2 = \|\mathbf{v}\|^2 \|\mathbf{w}\|^2$.
   (b) If $\mathbf{v} \cdot \mathbf{w} = 0$ and $\mathbf{v} \times \mathbf{w} = \mathbf{0}$, then $\mathbf{v} = \mathbf{0}$ or $\mathbf{w} = \mathbf{0}$.
   (c) If $\mathbf{v} \times \mathbf{w} = \mathbf{0}$ for all $\mathbf{w} \in \mathbb{R}^3$, then $\mathbf{v} = \mathbf{0}$.

7. Find the normal form of the plane containing the two lines
$$\frac{x-6}{4} = y + 3 = z, \qquad \text{and} \qquad \frac{x-11}{3} = \frac{y-14}{-6} = \frac{z+9}{2}.$$
Determine first whether there exists a plane containing both given lines. Justify your answer.

8. Find the line of intersection of the planes
$$x + 3y + 2z - 6 = 0, \qquad \text{and} \qquad 2x - y + z + 2 = 0.$$

9. Find the points of intersection of the sphere
$$(x-3)^2 + (y+1)^2 + (z-3)^2 = 9$$
and the line
$$x = -1 + 2t, \qquad y = -2 - 3t, \qquad z = 3 + t.$$

10. Find the intersection of the sphere
$$x^2 + y^2 + z^2 = 9$$
and the cylinder
$$x^2 + y^2 = 4.$$

11. Describe the intersection of the surfaces whose equations in spherical coordinates are
$$\theta = \frac{\pi}{2}, \qquad \phi = \frac{\pi}{4}.$$

12. Let
$$\mathbf{f}(t) = \left(\frac{\cos t}{\sqrt{1+a^2t^2}}, \frac{\sin t}{\sqrt{1+a^2t^2}}, \frac{-at}{\sqrt{1+a^2t^2}} \right),$$
where $a \neq 0$.
    (a) Show that $\|\mathbf{f}(t)\| = 1$ for all t.
    (b) Show directly that $\mathbf{f}'(t) \cdot \mathbf{f}(t) = 0$ for all t.

13. If $\mathbf{f}'(t) = \mathbf{0}$ for all t in some interval (a, b), show that $\mathbf{f}(t)$ is a constant vector on (a, b).

14. Show that
$$\frac{d}{dt}\left(\mathbf{f} \times \frac{d\mathbf{f}}{dt} \right) = \mathbf{f} \times \frac{d^2\mathbf{f}}{dt^2}.$$

15. The Mean Value Theorem does not hold for vector-valued functions. Show that for
$$\mathbf{f}(t) = (\cos t, \sin t, t),$$
there is no $t \in (0, 2\pi)$ such that
$$\mathbf{f}'(t) = \frac{\mathbf{f}(2\pi) - \mathbf{f}(0)}{2\pi - 0}.$$

16. Consider the curve
$$\mathbf{f}(t) = (2\cos 3t,\ 2\sin 3t,\ 2t^{3/2}), \qquad t \in [0, 1].$$
Calculate its arc length and parametrize the curve by arc length.

17. Show that the arc length L of a curve whose spherical coordinates are $\rho = \rho(t)$, $\theta = \theta(t)$ and $\phi = \phi(t)$, for $t \in [a, b]$, is given by
$$L = \int_a^b \sqrt{\rho'(t)^2 + \rho(t)^2 \sin^2\phi(t)\, \theta'(t)^2 + \rho(t)^2 \phi'(t)^2}\, dt.$$

18. Evaluate the following limits, or show that they do not exist:
    (a) $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2 - y^2}{x^2 + y^2}$, (b) $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy^2}{x^2 + y^4}$, (c) $\lim\limits_{(x,y)\to(0,2)} \dfrac{\sin(xy)}{x}$, (d) $\lim\limits_{(x,y)\to(0,0)} \cos\left(\dfrac{1}{xy}\right)$,
    (e) $\lim\limits_{(x,y)\to(0,\ln 2)} e^{x-2y}$, (f) $\lim\limits_{(x,y)\to(1,1)} \dfrac{\ln(1 + x^2y^2)}{e^y \sin x}$, (g) $\lim\limits_{(x,y)\to(0,0)} (x^2 + y^2)\cos\left(\dfrac{1}{xy}\right)$, (h) $\lim\limits_{(x,y)\to(\frac{1}{4},0)} \sqrt{\cos(2xy)}$.

19. Evaluate the following limits, where $\mathbf{p} = (x, y, z)$:
    (a) $\lim\limits_{\mathbf{p}\to(1,2,-1)} \left(\dfrac{1}{x} + \dfrac{1}{y} + \dfrac{1}{z}\right)$, (b) $\lim\limits_{\mathbf{p}\to(1,-1,2)} \dfrac{xy + 2yz}{x^2 + y^2 + z^2}$, (c) $\lim\limits_{\mathbf{p}\to(-\frac{1}{2},\frac{\pi}{4},2)} \tan^{-1}(xyz)$.

20. Given $\varepsilon > 0$, find a $\delta > 0$ such that

$$\sqrt{x^2 + y^2} < \delta \quad \Longrightarrow \quad |f(x, y) - f(0, 0)| < \varepsilon,$$

where

$$f(x, y) = \frac{2x^2 + xy + y^2}{1 + x^2 + y^2}, \qquad \varepsilon = 0.49.$$

21. Find all the second-order partial derivatives of the function

$$g(x, y) = x^2 e^{xy} + y^3 \sin x.$$

22. Use the limit definition of partial derivatives to compute $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ at the point $(1, -1)$, where

$$f(x, y) = \sqrt{5 + x^2 + xy + 2y^2}.$$

23. Find $\dfrac{\partial w}{\partial r}$ at $(r, s) = (1, 0)$, if

$$w = x^2 y + e^z, \qquad x = r + s^2, \qquad y = \cos(r - s), \qquad z = rs.$$

24. Find the directional derivative of the function

$$f(x, y, z) = \ln(1 + x^2 + 2y^2 + z^2) + xyz$$

at the point $P = (1, 1, 0)$ in the direction of the vector $\mathbf{u} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

25. Find all the critical points of the function

$$f(x, y) = x^3 + y^3 - 3x - 3y$$

and classify each of them as a local maximum, local minimum, or saddle point.

26. Use the substitution $r = \sqrt{x^2 + y^2}$ to show that

$$\lim_{(x,y)\to(0,0)} \frac{\sin\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}} = 1.$$

27. Let u and v be twice-differentiable functions of a single variable, and let $c \neq 0$ be a constant. Show that

$$f(x, y) = u(x + cy) + v(x - cy)$$

is a solution of the following one-dimensional wave equation

$$\frac{\partial^2 f}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 f}{\partial y^2} = 0.$$

28. Find the equation of the tangent plane to the surface

$$z = \ln(1 + x^2 + y^2) + \sin(xy)$$

at the point $P = (1, 0, \ln 2)$.

29. Find the equation of the tangent plane to the surface

$$x^2 + 2y^2 + 3z^2 + xy - yz = 6$$

at the point $P = (1, 1, 1)$.

30. Find the directional derivative of

$$f(x,y,z) = x^2 e^{yz} + \sin(xyz)$$

at the point $P = (1,1,1)$ in the direction of

$$\mathbf{v} = \frac{1}{\sqrt{14}}(1,2,3).$$

31. Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$ and $r(x,y) = \sqrt{x^2+y^2}$. Show that

$$\nabla r = \frac{1}{r}\mathbf{r}, \qquad (x,y) \neq (0,0),$$

and that

$$\nabla(r^2) = 2\mathbf{r}.$$

32. Find three positive numbers x, y, z whose sum is 12 such that x^2yz^2 is a maximum.

33. Evaluate the following integrals:

(a) $\displaystyle\int_1^4 \int_0^4 \left(\frac{x}{2} + \sqrt{y}\right) dx\,dy$, (b) $\displaystyle\int_0^{\ln 2} \int_0^{\ln 5} e^{2x+y}\,dx\,dy$, (c) $\displaystyle\int_{-1}^2 \int_0^{\pi/2} y\sin x\,dx\,dy$, (d) $\displaystyle\int_1^4 \int_1^e \frac{\ln x}{xy}\,dx\,dy$.

34. Find the volume bounded above by the paraboloid $z = x^2 + y^2$ and below by the square

$$R: \quad -1 \leq x \leq 1, \qquad -1 \leq y \leq 1.$$

35. Find the volume of the region bounded above by the surface $z = 4 - y^2$ and below by the square

$$R: \quad 0 \leq x \leq 1, \qquad 0 \leq y \leq 2.$$

36. Sketch the region of integration and evaluate the following integrals:

(a) $\displaystyle\int_0^\pi \int_0^x x\sin y\,dy\,dx$, (b) $\displaystyle\int_0^\pi \int_{\sin x}^1 y\,dy\,dx$, (c) $\displaystyle\int_1^4 \int_0^{\sqrt{x}} \frac{1}{2}e^{y/2}\,dy\,dx$.

37. Use a change in the order of integration to evaluate the following integrals:

(a) $\displaystyle\int_0^1 \int_1^{e^x} dy\,dx$, (b) $\displaystyle\int_0^{a\sqrt[3]{16}} \int_{\frac{x^2}{4a}}^{\sqrt{ax}} dy\,dx$ (with $a > 0$), (c) $\displaystyle\int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} 6x\,dy\,dx$,

(d) $\displaystyle\int_0^3 \int_1^{\sqrt{4-x}} (x+y)\,dy\,dx$, (e) $\displaystyle\int_0^1 \int_{x^3}^{\sqrt{x}} xy\,dy\,dx$, (f) $\displaystyle\int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} y\,dx\,dy$.

38. Evaluate the integral

$$\int_0^1 \int_0^{1-x} e^{\frac{y}{x+y}}\,dy\,dx,$$

using the transformation $x + y = u$ and $y = uv$.

39. Consider the transformation T from the xy-plane to the uv-plane given by

$$T: \quad x = \frac{1}{2}(u+v), \qquad y = \frac{1}{2}(u-v).$$

(a) Calculate the Jacobian of the transformation T.

(b) Evaluate

$$\iint_D (x-y)\cos^2(x+y)\,dA$$

using the transformation T, where D is the square in the xy-plane with vertices $(0,0), (\pi,\pi), (0,2\pi)$ and $(-\pi,\pi)$.

40. Evaluate the following integrals by making appropriate changes of variables:

(a) $\iint_D x^2\,dx\,dy$, where D is the elliptic region

$$\left\{(x,y) : \frac{x^2}{4} + \frac{y^2}{9} \leq 1\right\}.$$

(b) $\iint_D y^2\,dx\,dy$, where D is the region bounded by the curves

$$xy = 1, \quad xy = 2, \quad xy^2 = 1, \quad xy^2 = 2.$$

(c) $\iint_D (x+y)^2\,dx\,dy$, where D is the parallelogram bounded by the lines

$$x + y = 0, \quad x + y = 1, \quad 2x - y = 0, \quad 2x - y = 3.$$

41. Find the volume of the region bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$, using a double integral.

42. Find the area of the paraboloid

$$2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

inside the cylinder

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

43. Calculate the area of the region bounded by the upper half of the circle

$$x^2 + y^2 = 25,$$

the x-axis and the ordinates $x = -3$ and $x = 4$.

44. Find the area of the surface of the paraboloid

$$x^2 + y^2 = z,$$

which lies between the planes $z = 0$ and $z = 1$.

45. Write six different iterated triple integrals for the volume of the region in the first octant enclosed by the cylinder

$$x^2 + z^2 = 4$$

and the plane $y = 3$. Evaluate one of the integrals.

46. Write six different iterated triple integrals for the volume of the region bounded by the paraboloids

$$z = 8 - x^2 - y^2, \qquad z = x^2 + y^2.$$

Evaluate one of the integrals.

47. Evaluate the following integrals by changing the order of integration:

(a) $\displaystyle\int_0^4\int_0^1\int_{2y}^4 \frac{4\cos(x^2)}{2\sqrt{z}}\,dx\,dy\,dz$, (b) $\displaystyle\int_0^1\int_0^1\int_{x^2}^1 12xze^{y^2}\,dy\,dx\,dz$, (c) $\displaystyle\int_0^1\int_{\sqrt{z}}^1\int_0^{\ln 3} \frac{\pi e^{2x}\sin(\pi y^2)}{y^2}\,dx\,dy\,dz$.

48. Evaluate the following iterated triple integrals:

(a) $\displaystyle\int_0^1\int_0^1\int_0^1 (x^2+y^2+z^2)\,dz\,dy\,dx$, (b) $\displaystyle\int_0^{\pi/6}\int_0^1\int_{-2}^3 y\sin z\,dx\,dy\,dz$.

(c) $\displaystyle\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}}\,dz\,dy\,dx$.

49. Evaluate the following integrals using cylindrical coordinates:

(a) $\displaystyle\int_0^5\int_0^{\sqrt{25-x^2}}\int_0^6 \frac{1}{\sqrt{z^2+x^2+y^2}}\,dz\,dy\,dx$.

(b) $\displaystyle\int_0^2\int_0^{\sqrt{4-x^2}}\int_0^{\frac{x^2+y^2}{2}} \frac{z}{\sqrt{z^2+x^2+y^2}}\,dz\,dy\,dx$.

(c) $\displaystyle\iiint_V \sqrt{x^2+y^2}\,dV$, where V is the solid in the first octant bounded by the coordinate planes, the plane $z=4$, and the cylinder $x^2+y^2=25$.

(d) Evaluate the integral $\displaystyle\iiint_V (x^2+y^2)^{3/2}\,dV$, where V is the solid bounded above by the xy-plane, below by the plane $z=-2$, and laterally by the cylinder $x^2+y^2=4$.

50. Evaluate the following integrals using spherical coordinates:

(a) $\displaystyle\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{x^2+y^2+z^2}}\,dz\,dy\,dx$.

(b) $\displaystyle\int_0^3\int_0^{\sqrt{9-x^2}}\int_0^{\sqrt{9-x^2-y^2}} xz\,dz\,dy\,dx$.

(c) $\displaystyle\iiint_V \sin\sqrt{x^2+y^2+z^2}\,dV$, where V lies inside the sphere $x^2+y^2+z^2=49$ and above the cone $z=\sqrt{x^2+y^2}$.

51. Find the volume V inside the sphere $x^2+y^2+z^2=1$ and above the cone $z=\sqrt{x^2+y^2}$.

52. Show that the volume inside the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ is $\dfrac{4\pi abc}{3}$.

53. Find the volume inside the elliptic cylinder

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, \qquad 0\le z\le 2.$$

54. Find the center of mass of the following regions with the given density functions:

(a) $R=\{(x,y): y\ge 0,\ x\ge 0,\ 1\le x^2+y^2\le 4\}$, with density function $\delta(x,y)=\sqrt{x^2+y^2}$.

(b) $R=\{(x,y): y\ge 0,\ x^2+y^2\le 1\}$, with density function $\delta(x,y)=y$.

55. Find the center of mass of the following solids with the given density functions:

(a) $S=\{(x,y,z): z\ge 0,\ x^2+y^2+z^2\le a^2\}$, with density function $\delta(x,y,z)=x^2+y^2+z^2$.

(b) $S=\{(x,y,z): x\ge 0,\ y\ge 0,\ z\ge 0,\ x+y+z\le 1\}$, with density function $\delta(x,y,z)=1$.